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## 5.8 SETTING SYSTEM MONITOR VALUES

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### INITIAL SETUP:

**Tools and Special Tools**

None

**References**

5 System Configuration WP  
6.2 Calibration WP

**Materials/Parts**

None

**Equipment Condition**

Tilt Sensor Configuration WP Completed  
Equipment Powered On  
Generator or Shore power applied

**Personnel Required**

Technician (2)

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### GENERAL INFORMATION

- This procedure is designed to set monitor limits for each of system's functional assemblies. This includes three sensors, interrogation transmitter and each GTU. Each must be set before system can be put into an operational mode.
- MultiBIT tool is used to automatically calculate monitor limits.
- The following procedure will need to be repeated for the second rack if system is configured with dual racks.
- Switch to the rack being configured on the Remote Control Unit (RCU).

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## Setting System Monitor Limits

### Continuous BIT Procedure

#### NOTE

If system has been operating and RAM files from previous day have not been archived, do so at this time before configuring Cal/BIT monitor nominal values.

1. Stop the system software from the MI.
2. Run an Archive as follows:
  - a. Click "Archive."
  - b. Click "Run" in the dialog box that opens.
  - c. Once the dialog box shows "Done" click close.
3. Start the system software.
  - a. Click System.
  - b. Select "Test Mode."

- c. Check "Enable" under periodic BIT.
- d. Set interval to "0."
- e. Click "Apply Settings."
- f. Allow BIT to run for a minimum of 10 cycles.
- g. Stop continuous BIT by unchecking "Enable" and clicking "Apply Settings" when visible.
- h. Close "Test Mode."
- i. Click "Stop", when system mode indicates Maintenance or Track.

## **ASA, ATA, ESA, Interrogator and Uplink Monitor Values**

### **NOTE**

- This procedure sets monitor values for ESA, ASA and ATA sensor assemblies.
- This procedure requires first performing the Command BIT procedure.

1. Click Review BIT Monitor Data (MultiBIT) in Tools menu to open MultiBIT tool in MI.
2. Close the MI.
3. Click Open ASA BIT in File Menu of MultiBIT.

### **NOTE**

- Tool will display current settings and computed values. Color coding is used to show where current settings are beyond limits.
- Red indicates current setting is beyond alarm limits and yellow indicates it is beyond alert limits (See Figure 8-1).
- Clicking Save will save monitor limits to both bases.

4. Click Compute Nominals in Tools menu of MultiBIT,

| Parameter     | Current  | Computed | Delta | Std Dev | Units  | Save                                |
|---------------|----------|----------|-------|---------|--------|-------------------------------------|
| TOA Near      | 5746.7   | 5747.0   | 0.3   | 0.000   | ns     | <input checked="" type="checkbox"/> |
| TOA Far       | 395746.7 | 395747.0 | 0.3   | 0.000   | ns     | <input checked="" type="checkbox"/> |
| Amp Ref       | 1750     | 1515     | -235  | 0.000   | mV     | <input checked="" type="checkbox"/> |
| Amp Low       | 1744     | 1507     | -237  | 0.000   | mV     | <input checked="" type="checkbox"/> |
| Amp Medium    |          |          |       |         |        |                                     |
| Amp High      | 1620     | 1484     | -136  | 0.000   | mV     | <input checked="" type="checkbox"/> |
| Phase L       | 0        | 0        | 0     | 0.000   | counts | <input checked="" type="checkbox"/> |
| Phase M       |          |          |       |         |        |                                     |
| Phase H       | 0        | 0        | 0     | 0.000   | counts | <input checked="" type="checkbox"/> |
| Delta Phase L | 0        | 0        | 0     | 0.000   | counts | <input checked="" type="checkbox"/> |
| Delta Phase M |          |          |       |         |        |                                     |
| Delta Phase H | 0        | 0        | 0     | 0.000   | counts | <input checked="" type="checkbox"/> |
| Frequency     | 1090.000 | 1090.018 | 0.018 | 0.000   | MHz    | <input type="checkbox"/>            |
| Tilt Para     |          |          |       |         |        |                                     |
| Tilt Perp     |          |          |       |         |        |                                     |

Data Path: \\base2\\c\\run

☐ Show Current Limits ☐ Show New Limits

Less << Save Close

**Figure 8-1. ASA Computed Nominals**

- Click Save in Computed Nominals screen of MultiBIT (See Figure 8-1).
- Click Open ATA BIT in File Menu of MultiBIT.

#### NOTE

- Tool will display current settings and computed values. Color coding is used to show where current settings are beyond limits.
- Red indicates current setting is beyond alarm limits and yellow indicates it is beyond alert limits (See Figure 8-2).
- Clicking Save will save monitor limits to both bases.

- Click Compute Nominals in Tools menu of MultiBIT,

| Parameter     | Current  | Computed | Delta | Std Dev | Units | Save                                |
|---------------|----------|----------|-------|---------|-------|-------------------------------------|
| TOA Near      | 6579.7   | 6576.0   | -3.7  | 0.000   | ns    | <input checked="" type="checkbox"/> |
| TOA Far       | 396579.7 | 396576.0 | -3.7  | 0.000   | ns    | <input checked="" type="checkbox"/> |
| Amp Ref       |          |          |       |         |       |                                     |
| Amp Low       |          |          |       |         |       |                                     |
| Amp Medium    | 1546     | 1304     | -242  | 0.000   | mV    | <input checked="" type="checkbox"/> |
| Amp High      |          |          |       |         |       |                                     |
| Phase L       |          |          |       |         |       |                                     |
| Phase M       |          |          |       |         |       |                                     |
| Phase H       |          |          |       |         |       |                                     |
| Delta Phase L |          |          |       |         |       |                                     |
| Delta Phase M |          |          |       |         |       |                                     |
| Delta Phase H |          |          |       |         |       |                                     |
| Frequency     |          |          |       |         |       |                                     |
| Tilt Para     |          |          |       |         |       |                                     |
| Tilt Perp     |          |          |       |         |       |                                     |

Data Path: \\base2\\c\\run

☐ Show Current Limits ☐ Show New Limits

Less << Save Close

**Figure 8-2. ATA Computed Nominals**

8. Click Save in Computed Nominals screen of MultiBIT (See Figure 8-2).
9. Click "Open ESA BIT" in File menu of MultiBIT.

**NOTE**

- Tool will display current settings and computed values. Color coding is used to show where current settings are beyond limits.
- Red indicates current setting is beyond alarm limits and yellow indicates it is beyond alert limits (See Figure 8-3).
- Clicking Save will save monitor limits to both bases.

10. Click Compute Nominals in Tools menu of MultiBIT.

| Parameter     | Current  | Computed | Delta | Std Dev | Units   | Save                                |
|---------------|----------|----------|-------|---------|---------|-------------------------------------|
| TOA Near      | 6421.9   | 6421.0   | -0.9  | 0.000   | ns      | <input checked="" type="checkbox"/> |
| TOA Far       | 396421.9 | 396421.0 | -0.9  | 0.000   | ns      | <input checked="" type="checkbox"/> |
| Amp Ref       | 1650     | 1414     | -236  | 0.000   | mV      | <input checked="" type="checkbox"/> |
| Amp Low       | 1474     | 1234     | -240  | 0.000   | mV      | <input checked="" type="checkbox"/> |
| Amp Medium    | 1509     | 1390     | -119  | 0.000   | mV      | <input checked="" type="checkbox"/> |
| Amp High      | 1420     | 1179     | -241  | 0.000   | mV      | <input checked="" type="checkbox"/> |
| Phase L       | 0        | 0        | 0     | 0.000   | counts  | <input checked="" type="checkbox"/> |
| Phase M       | 0        | 0        | 0     | 0.000   | counts  | <input checked="" type="checkbox"/> |
| Phase H       | 0        | 0        | 0     | 0.000   | counts  | <input checked="" type="checkbox"/> |
| Delta Phase L | 0        | 0        | 0     | 0.000   | counts  | <input checked="" type="checkbox"/> |
| Delta Phase M | 0        | 0        | 0     | 0.000   | counts  | <input checked="" type="checkbox"/> |
| Delta Phase H | 0        | 0        | 0     | 0.000   | counts  | <input checked="" type="checkbox"/> |
| Frequency     | 1090.000 | 1090.018 | 0.018 | 0.000   | MHz     | <input type="checkbox"/>            |
| Tilt Para     | 0.000    | 0.000    | 0.000 | 0.000   | degrees | <input type="checkbox"/>            |
| Tilt Perp     | 0.000    | 0.000    | 0.000 | 0.000   | degrees | <input type="checkbox"/>            |

Data Path: \\base2\c\run

☐ Show Current Limits    ☐ Show New Limits

Less <<    Save    Close

**Figure 8-3. ESA Computed Nominals**

11. Ensure Tilt sensor Save check boxes are selected.
12. Click Save in Computed Nominals screen of MultiBIT (See Figure 8-3).

#### NOTE

This procedure sets monitor values for interrogator assembly. This procedure requires first performing Command BIT procedure.

13. Click Open Interrogator BIT in File menu of MULTIBIT (See Figure 8-4).

**NOTE**

- Tool will display current settings and computed values. Color coding is used to show where current settings are beyond limits.
- Red indicates current setting is beyond alarm limits and yellow indicates it is beyond alert limits (See Figure 8-4).
- Clicking Save will save monitor limits to both bases.

14. Click Compute Nominals in Tools menu of MultiBIT,

| Parameter | Current | Computed | Delta | Std Dev | Units | Save                                |
|-----------|---------|----------|-------|---------|-------|-------------------------------------|
| P1/P3 Amp | 550     | 531      | -19   | 0.000   | mV    | <input checked="" type="checkbox"/> |
| P2 Amp    | 550     | 531      | -19   | 0.000   | mV    | <input checked="" type="checkbox"/> |

Data Path: c:\docume~1\admini~1\locals~1\temp\multibitftp

☐ Show Current Limits ☐ Show New Limits

Save Close

**Figure 8-4. ATA Interrogation Transmitter Compute Nominal Values**

15. Click Save in Computed Nominals screen of MultiBIT (See Figure 8-4).

16. Click Open Uplink\_1 BIT in File menu of MultiBIT.

**NOTE**

- Tool will display current settings and computed values. Color coding is used to show where current settings are beyond limits.
- Red indicates current setting is beyond alarm limits and yellow indicates it is beyond alert limits (See Figure 8-3).

17. Click Compute Nominals in Tools menu of MultiBIT,

18. Click Save in Computed Nominals screen of MultiBIT.

19. Repeat steps for Uplink\_2, Uplink\_3 and Uplink\_4 if they exist (up to four GTUs may be installed)

**NOTE**

From \\base2\C\Run\BIT file path selects the remaining Uplinks.

20. To open remaining Uplink BIT files, click on the "Open Folder."

21. Change file type to "Uplink BIT files."

**NOTE**

Do not change the file name.

22. Click Save in Computed Nominal screen of MultiBIT.
23. Click “Yes” when prompted on the Confirm Save As dialog box.
24. Click “Yes” when prompted on the MultiBIT dialog box.
25. Click ‘OK’.
26. Return to the desktop and open Send to Bases icon.
27. Check monitor\_limits.dat.
28. Select browse associated to monitor\_limits.dat.
29. Select the monitor\_limits.dat file.
30. Select Open.
31. Ensure that both BASE1 and BASE2 are checked.
32. Click “Send”.
33. Click “Exit” once Send to Base is completed.

**Validation Nominal Monitor Values****NOTE**

This procedure is designed to help technician validate that monitor values that have been set are correct and will allow system to be brought up into an operational mode.

1. Open the MI.
2. Perform Archive.
3. Click “Start” to start system software in MI, wait for system to complete “Diagnostic” mod
4. Enter either Track or Maintenance Mode.
5. Click “Test Mode” in System Menu of MI.
6. Click “Command BIT” and wait for BIT to complete from Test Mode Dialog.
7. Repeat a minimum of 5 times.
8. Click “Close” to exit Test Mode.
9. Click “SYSTEM STATUS” to open System Status Window.
10. Select “History” and wait for screen to update under View.

**NOTE**

- Any LRUs that are not installed will display in blue.
- If any of status indicators are yellow or red, click on specific button to see what alerts/alarms have been recorded for LRU.
- Go back through set up procedure for that indicated nominal for the LRU.

11. Verify all LRU status indicators are green.

**END OF TASK**

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